

Lightweight Medical Image Segmentation via Cross-Image Region Contrastive and Global Boundary Distillation

Linxi Tang¹[0009-0004-5330-8671], Wanli Dong¹(✉)[0000-0003-2680-3736],
Xiaoming Gao¹[0000-0002-2818-1400], Fudie Ai¹[0009-0003-2592-9836], Zhuoling
Xie¹[0009-0002-8802-9221], and Lina Tang¹[0009-0003-6909-7621]

Southwest University of Science and Technology, Mianyang 621010, Sichuan, China
wldong81@foxmail.com

Abstract. Advances in deep learning have led to highly accurate yet parameter-heavy medical image segmentation models, which pose challenges for practical deployment. Knowledge distillation has emerged as a promising approach for model compression; however, existing methods face two key challenges in medical applications: (1) high intra-class shape variability and low inter-class contrast among anatomical structures, and (2) severe class imbalance due to drastic size differences between organs. To address these challenges, we propose a novel knowledge distillation framework. It consists of two complementary modules: (i) Cross-Image Region Contrastive Distillation (CIRD), which enhances feature discriminability by maximizing intra-class similarity and minimizing inter-class similarity across regions in a batch; and (ii) Global Boundary Distillation (GBD), which extracts global boundary masks and assigns uniform importance to all organ boundaries, effectively mitigating bias toward large organs. Extensive experiments on three benchmark datasets—BTCV, Synapse, FLARE—demonstrate that CIRGBD consistently improves lightweight student performance, achieving up to 4.43% Dice score gain over baseline distillation methods.

Keywords: Medical Image Segmentation · Knowledge distillation · Category distinction · Boundary alignment.

1 Introduction

Medical image segmentation is a challenging task in computer vision. Recent advances in deep learning models have achieved high accuracy in this task, but they are often computationally expensive and parameter-heavy, making deployment in real-world clinical settings challenging. Knowledge distillation addresses this issue by transferring knowledge from a high-capacity teacher model to a lightweight student model, enabling the student to achieve competitive performance while significantly reducing model size and computational cost. Consequently, distillation has become an essential strategy for compressing segmentation models without severely compromising accuracy.

For example, Liu et al. propose the SKD [1] method, which represents knowledge in semantic segmentation by constructing structural relationships. IFVD [2] extracts knowledge based on semantic masks, while CWD [3] achieves knowledge distillation by enhancing the alignment of salient regions between teacher and student models across channels. However, these approaches primarily focus on spatial information, overlooking the importance of intra-class and inter-class information, as well as boundary alignment. While EMKD [4] and CIRKD [5] incorporate contrastive learning, their application is confined to intra-class comparisons within a single image, and they do not explicitly model boundaries. BPKD [6] incorporates category-specific boundary alignment but shows limited performance beyond large-category datasets such as CityScapes [7]. Particularly in the context of medical image segmentation, these methods face two major challenges: (1) substantial morphological and textural variations within the same organ due to differences in imaging angles and scales, which make spatial information alone insufficient for accurate category discrimination; and (2) severe organ size imbalance, where boundary pixels of large organs dominate the distillation process during category boundary alignment, resulting in blurred boundaries for smaller organs. To address these issues, we propose a novel distillation framework

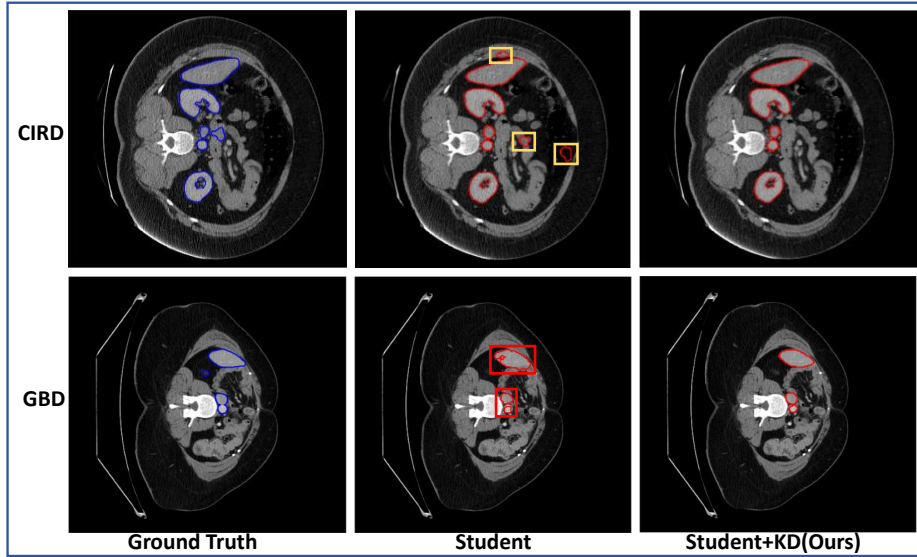


Fig. 1. Visualization of CIRD and GBD. Using the BTCV [8] dataset as an example, red and yellow boxes are highlighted. The yellow boxes indicate regions with poor category differentiation, which are effectively corrected by our CIRD method. The red boxes denote blurred and uneven boundaries, where our GBD method successfully produces accurate and well-defined boundary segmentation.

for medical image segmentation. Specifically, to address the first challenge, we introduce Cross-Image Region Contrastive Distillation (CIRD), which extracts category region features from the intermediate layers and performs category-level

contrast between images, thereby learning intra-class similarities and inter-class differences. This facilitates accurate segmentation of organs with intra-class morphological variations while enhancing category discrimination. Regarding the second challenge, we propose Global Boundary Distillation (GBD), in which global boundary masks are extracted from the teacher model, and global boundary alignment is applied on the final score layer for each category channel. This approach assigns equal importance to all organ boundaries, mitigates dominance by large organs, and effectively transfers the boundary handling capability of the teacher model. The effectiveness of the proposed modules is illustrated in Figure 1. The main contributions of this work are summarized as follows.

- We propose a novel medical image segmentation distillation framework integrating Cross-Image Region Contrastive Distillation (CIRD) and Global Boundary Distillation (GBD), which effectively addresses the intra-class shape variability, low inter-class contrast and severe class imbalance in medical imaging, and significantly enhances the segmentation performance of lightweight models.
- We conduct extensive experiments on three benchmark medical datasets (BTCV, Synapse, FLARE), and the proposed method achieves up to 4.43% Dice score gain over baseline distillation methods, demonstrating superior performance in medical image segmentation distillation.

2 Related Work

Medical Image Segmentation. U-Net[9] is a benchmark for medical image segmentation, using an encoder-decoder structure and skip connections to fuse spatial and semantic features. Subsequent advancements (RESUNet[10], Residual-Attention UNet++[11], RA-UNET[12]) enhance feature representation, boundary localization, and adaptability to anatomical variations, serving as high-performance teacher models. Lightweight design is critical for clinical deployment. To address high parameters and computational costs, schemes like ENET[13] (channel pruning, depthwise convolutions) and RESNET18-UNET[10] (lightweight residual variants) balance accuracy and efficiency.

Edge Detection. Classical edge detection algorithms like Canny[14], Sobel[15], and Prewitt[16] have been retrofitted into modern deep learning architectures to achieve fine-grained segmentation. Deeply-supervised edge detection methods such as HED[17] and RCF[18] introduce multi-scale edge information directly into the segmentation pipelines. Similarly, models like CASENet[19] have advanced the state-of-the-art by fusing class-specific edges into segmentation algorithms, providing a dual benefit of detailed boundary representation and class differentiation. In parallel, techniques like edge-attention models incorporate edge information by weighting features based on their boundary importance, enabling finer contour mapping in semantic segmentation.

Knowledge Distillation. Knowledge Distillation (KD) aims to transfer knowledge from one or more complex teacher models to lightweight student models, which is mainly applied to basic computer vision tasks and divided into

three paradigms: response-based, feature-based, and relation-based. Traditional KD is mostly tailored for image classification tasks and has limited applicability in pixel-level segmentation. In recent years, specialized KD methods for semantic segmentation have emerged, including structural knowledge distillation, channel-wise distillation (CWD)[3], Boundary-Preserving Knowledge Distillation (BPKD)[6], etc. In addition, there are techniques such as intra-class feature variation distillation and cross-image relational distillation. Relevant experiments show that these methods can effectively improve segmentation performance.

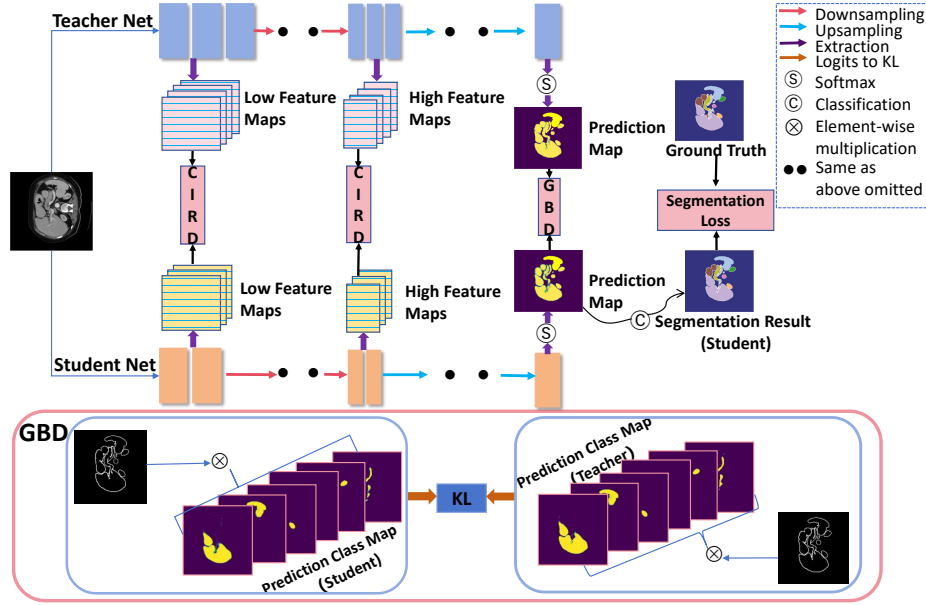


Fig. 2. Overall framework and GBD architecture diagram. From left to right, CIRD extracts regional information from both low-level and high-level feature layers, while GBD performs global boundary alignment at the prediction layer. Finally, the output is combined with the baseline cross-entropy segmentation distillation loss.

3 Methodology

The overall architecture proposed in this paper is illustrated in Figure 2, comprising the CIRD and GBD modules.

In this work, low-level features correspond to early encoder layers that retain rich spatial details, such as organ boundaries and texture patterns, whereas high-level features are extracted from deeper layers and encode more abstract semantic and structural information. The combination of multi-level features enables CIRD to capture robust region-level representations that are well suited for medical image segmentation.

3.1 Cross-Image Region Contrastive Distillation

CIRD is designed to model cross-image global structural relationships rather than fine-grained spatial alignment. Unlike pixel-level alignment methods, CIRD focuses on learning semantically consistent region representations across images, which are more robust to intra-class shape and appearance variations commonly observed in medical data. Therefore, average pooling is adopted to obtain stable and semantically consistent region-level representations. The learning of fine-grained spatial structures is primarily enforced by the pixel-wise segmentation loss, making the two objectives complementary and mitigating potential information loss caused by regional averaging.

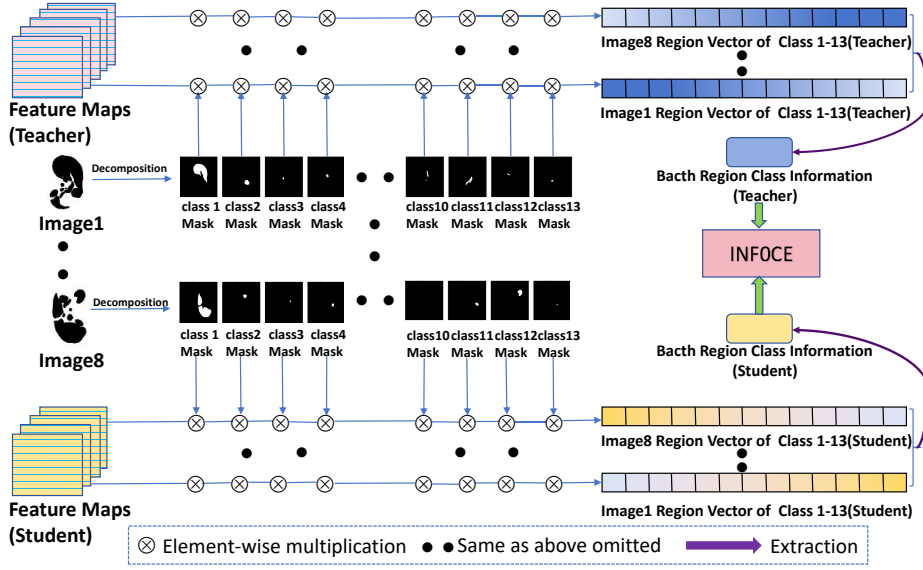


Fig. 3. CIRD architecture diagram. The CIRD module is illustrated using the BTCV dataset with a batch size of 8 as an example. It simultaneously takes feature maps from both the teacher and student models as inputs. The feature vectors for each category region across the eight CT images are obtained by applying the corresponding organ category masks to each image in the batch. Then, both the teacher’s and student’s batch region category feature information are passed to the INFOCE module, which enforces intra-class similarity while maximizing inter-class dissimilarity.

CIRD effectively distinguishes missegmented regions, as illustrated in the upper part of Figure 1. Figure 3 illustrates the structure of CIRD. Specifically, it extracts binarized class masks from the teacher model’s prediction map, resizes them to match the feature map, and multiplies each class mask with all channels to obtain class-specific weighted features. Average pooling over the class regions then produces the feature vector R for each class. The formula is as follows:

$$R_c = \frac{\sum_{h=1}^H \sum_{w=1}^W (x \cdot \hat{m}_c)_{h,w}}{\sum_{h=1}^H \sum_{w=1}^W \hat{m}_c} \quad (1)$$

where $x \in \mathbb{R}^{B \times C_f \times H \times W}$ denotes the feature map, C_f is the number of feature channels, and $\hat{m}_c \in \mathbb{R}^{B \times 1 \times H \times W}$ represents the class mask for class c in the batch. $R_c \in \mathbb{R}^{B \times C_f}$ corresponds to the class-specific features within the batch. Specifically, the first row of R_c contains the feature vector for class c in the first image of the batch, the second row corresponds to the second image, and so on. This formulation enables cross-image region comparison, which is key to enhancing the discriminability of intra-class features while preserving inter-class distinctions.

Within the batch, the features of each category region are stacked along the category dimension to form a tensor containing the features of all categories: $\mathbf{F} = [\mathbf{R}_1, \mathbf{R}_2, \dots, \mathbf{R}_{C_c}] \in \mathbb{R}^{B \times C_c \times C_f}$, where C_c denotes the number of categories. The extracted cross-image region features from the student (F_s) and teacher (F_t) models are then used to perform contrastive learning via the InfoNCE module, which encourages higher similarity for features of the same class while reducing similarity for features of different classes.

Specifically, L2 normalization is first applied to both the student features F_s and teacher features F_t , after which they are flattened along the category dimension to obtain 2D matrices of shape $[B \times C_c, C_f]$, resulting in $s_{\text{vectors}} = \text{reshape}(F_s, [-1, C_f])$, $t_{\text{vectors}} = \text{reshape}(F_t, [-1, C_f])$. Subsequently, the similarity between student and teacher category features is computed as follows:

$$S = \frac{s_{\text{vectors}} \cdot t_{\text{vectors}}^T}{\tau} \quad (2)$$

where S is the computed similarity matrix, $s_{\text{vectors}} \cdot t_{\text{vectors}}^T$ denotes the category similarity using the dot product (the greater the feature similarity, the larger the value), t^T vectors is the transpose of the teacher's feature vector. Finally, the final contrastive loss is computed using the cross-entropy loss. As follows:

$$L_{\text{CIRD}} = \frac{1}{B \times C} \sum_{i=1}^{B \times C} \left(-\log \left(\frac{e^{S_{i,i}/\tau}}{\sum_{j=1}^C e^{S_{i,j}/\tau}} \right) \right) \quad (3)$$

where τ denotes the temperature parameter that regulates the smoothness of the similarity distribution. The diagonal elements $S_{i,i}$ of the matrix S represent the similarity between student and teacher features of the same class within the batch, while the off-diagonal elements $S_{i,j}$ correspond to the similarity between features of different classes. After softmax normalization, the similarity of same-class features is amplified, guiding the student model to align with the teacher model. Conversely, the similarity between different-class features is relatively low and further suppressed by the softmax, enhancing the discriminability between classes. Employing cross-entropy loss effectively aligns same-class features while significantly increasing the distinction among different-class features.

3.2 Global Boundary Distillation

The conventional BPKD method aligns category-specific boundaries independently and performs well on datasets where inter-class size differences are relatively small. However, in medical image segmentation, substantial variations in

organ sizes often cause the distillation process to be dominated by large organs. Even when dynamic weighting strategies are introduced to emphasize categories with fewer boundary pixels, the resulting segmentation performance remains suboptimal.

To address this limitation, we propose a Global Boundary Distillation (GBD) module, as illustrated at the bottom of Fig. 2. The qualitative visualization shown in the lower part of Fig. 1 demonstrates that the proposed GBD effectively produces precise and smooth organ boundary delineations.

Specifically, we first apply softmax to the final prediction layers of both the student and teacher models along the channel dimension to obtain category-wise prediction score maps. We then employ the Sobel operator [15] to extract a global boundary mask $\text{mask}_{\text{global}}$ from the teacher’s prediction map. A threshold k is used to control the sensitivity of boundary detection, thereby determining which pixels are classified as boundary pixels. Subsequently, each category prediction channel is multiplied by the global boundary mask, and a Kullback–Leibler (KL) divergence is used to encourage the student model to align with the teacher’s boundary modeling capability. Although the global boundary mask is shared across categories, category-specific boundary information is implicitly preserved through the channel-wise prediction scores, as detailed below:

$$p = \sum_{h=1}^H \sum_{w=1}^W (x \cdot \text{mask}_{\text{global}})_{h,w}, \quad kl_{\text{div}} = \sum_{i=1}^C (p_i^s \parallel p_i^t), \quad (4)$$

where $\text{mask}_{\text{global}} \in \mathbb{R}^{B \times 1 \times H \times W}$ and $x \in \mathbb{R}^{B \times C \times H \times W}$ denote the global boundary mask and the category prediction score map, respectively, with C being the number of categories. p_i^s and p_i^t represent the student and teacher probabilities of the i -th category channel after being weighted by the global boundary mask.

The final global boundary loss is computed by normalizing the KL divergence with respect to the total number of effective boundary pixels:

$$L_{\text{global}} = \frac{kl_{\text{div}}}{\sum_{h=1}^H \sum_{w=1}^W \text{mask}_{\text{global}}}. \quad (5)$$

Although the global boundary mask does not explicitly distinguish category labels in the spatial domain, category-specific information is implicitly preserved through the channel-wise prediction scores. After softmax normalization, each category channel exhibits high confidence primarily around its corresponding boundary regions, while responses on boundaries of other categories remain low. Consequently, when aggregating boundary information using the global boundary mask, the dominant contribution to the KL divergence for each channel still comes from its own class boundaries, and the interference from other categories is effectively limited.

Moreover, we intentionally adopt a global boundary formulation rather than strictly category-specific boundary alignment. This design alleviates the severe organ size imbalance commonly observed in medical image segmentation, where large organs would otherwise dominate the distillation process. By assigning

equal importance to all boundary pixels, the proposed GBD encourages balanced boundary learning across categories while maintaining sufficient category discriminability through channel-wise supervision, leading to clearer and more discriminative boundary contours in the final segmentation results.

3.3 Overall Loss

We combine the modules mentioned above to train the student network using the formula shown in the end. The total loss function is as follows:

$$L_{\text{total}} = L_{\text{seg}} + \alpha_1 L_{\text{circlow}} + \alpha_2 L_{\text{circhigh}} + \beta L_{\text{global}} \quad (6)$$

where L_{seg} is the general cross-entropy loss function. The hyperparameters α_1 , α_2 , and β are set to 0.1, 0.5, and 0.1, respectively. The temperature parameter τ was set to 0.1, and the k threshold was set to 0.5.

4 EXPERIMENTS

4.1 Experimental Setup

We used standard data augmentation methods, including random flipping, rotation, cropping, etc. We conducted experiments on three medical image datasets: BTCV (13 classes) [8], Synapse (8 classes) [20], and Flare (4 classes) [21]. All experiments used Adam for optimization, with momentum parameters set to $\beta_1 = 0.9$, $\beta_2 = 0.999$, and a batch size of 8. The initial learning rate was 0.001, and training was performed for 80 epochs. All training methods used the same configuration on four 4090 (24GB) GPUs.

4.2 Evaluation Metric

In medical image segmentation, we evaluated using Dice [23] and mIoU [24]. The Dice coefficient is defined as:

$$\text{Dice}(P, G) = \frac{2 \sum_{c=1}^C |P_c \cap G_c|}{\sum_{c=1}^C |P_c| + |G_c|} \quad (7)$$

The mIoU metric is defined as:

$$\text{mIoU}(P, G) = \frac{1}{C} \sum_{c=1}^C \frac{|P_c \cap G_c|}{|P_c \cup G_c|} \quad (8)$$

where C represents the number of classes, P_c and G_c represent the predicted and ground truth regions of class c , respectively. To ensure fairness, we computed the average of the sum of Dice [23] and mIoU [24] in all classes as the performance metric.

Table 1. Compared to other methods on three datasets.

Methods	Param(M)	FLOPS(G)	FPS(S)	BTCV 512x512 [8]		Synapse 512x512 [20]		Flare 512x512 [21]	
				Dice(%)	mIoU(%)	Dice(%)	mIoU(%)	Dice(%)	mIoU(%)
T: RA-UNET [9,12]	22.14	78.22	27.42	86.17	83.11	92.76	90.03	87.00	84.00
S: ENET [13]	0.35	2.01	57.14	83.40	80.03	90.17	87.73	84.75	81.80
EMKD [4]	0.35	2.01	57.12	84.74(▲ 1.34)	81.61(▲ 1.58)	90.81(▲ 0.64)	87.87(▲ 0.14)	85.88(▲ 1.13)	82.80(▲ 1.00)
CIRKD [5]	0.35	2.01	56.42	84.58(▲ 1.18)	81.35(▲ 1.32)	90.23(▲ 0.06)	87.49(▼ 0.24)	86.27(▲ 1.52)	83.21(▲ 1.41)
CWD [3]	0.35	2.01	57.15	83.67(▲ 0.27)	80.27(▲ 0.24)	88.86(▼ 1.31)	85.10(▼ 2.63)	85.11(▲ 0.36)	81.97(▲ 0.17)
BPKD [6]	0.35	2.01	57.31	84.13(▲ 0.73)	80.64(▲ 0.61)	89.01(▼ 1.16)	85.56(▼ 2.17)	86.24(▲ 1.49)	82.96(▲ 1.16)
CIRGBD(Ours)	0.35	2.01	57.46	86.53(▲ 3.13)	83.42(▲ 3.39)	92.27(▲ 2.10)	89.39(▲ 1.66)	86.98(▲ 2.23)	83.97(▲ 2.17)
T: RA-UNET [9,12]	22.14	78.22	27.42	86.17	83.11	92.76	90.03	87.00	84.00
S: RESNET18-UNET [10]	10.92	55.31	28.26	85.04	82.10	91.17	88.26	83.27	80.19
EMKD [4]	10.92	55.31	28.72	85.98(▲ 0.94)	83.35(▲ 1.25)	92.48(▲ 1.31)	89.68(▲ 0.94)	85.11(▲ 1.84)	82.25(▲ 2.06)
CIRKD [5]	10.92	55.31	25.13	87.32(▲ 2.28)	84.32(▲ 2.22)	92.91(▲ 1.74)	90.26(▲ 2.00)	86.10(▲ 2.83)	83.18(▲ 2.99)
CWD [3]	10.92	55.31	28.79	87.62(▲ 2.58)	84.44(▲ 2.34)	93.20(▲ 2.03)	90.44(▲ 2.18)	86.17(▲ 2.90)	83.04(▲ 2.85)
BPKD [6]	10.92	55.31	27.89	86.26(▲ 1.22)	83.12(▲ 1.02)	92.13(▲ 0.96)	89.17(▲ 0.91)	84.95(▲ 1.68)	81.75(▲ 1.56)
CIRGBD(Ours)	10.92	55.31	29.31	88.00(▲ 2.96)	85.01(▲ 2.91)	93.53(▲ 2.36)	90.89(▲ 2.63)	86.41(▲ 3.14)	83.29(▲ 3.10)
T: RA-UNET [9,12]	22.14	78.22	27.42	86.17	83.11	92.76	90.03	87.00	84.00
S: RESNET18 [22]	11.18	9.52	47.62	77.92	73.84	89.93	85.56	81.35	76.58
EMKD [4]	11.18	9.52	47.12	80.60(▲ 2.68)	76.47(▲ 2.63)	89.42(▼ 0.51)	84.98(▼ 0.58)	82.79(▲ 1.44)	77.95(▲ 1.37)
CIRKD [5]	11.18	9.52	46.62	80.35(▲ 2.43)	76.29(▲ 2.45)	90.37(▼ 0.44)	86.13(▼ 0.57)	82.17(▲ 0.82)	77.72(▲ 1.14)
CWD [3]	11.18	9.52	47.91	78.04(▲ 0.12)	73.95(▲ 0.11)	87.55(▼ 2.38)	82.64(▼ 2.92)	81.89(▲ 0.54)	76.71(▲ 0.13)
BPKD [6]	11.18	9.52	47.86	78.85(▲ 0.93)	74.60(▲ 0.76)	89.27(▼ 0.66)	84.59(▼ 0.97)	82.33(▲ 0.98)	77.43(▲ 0.85)
CIRGBD(Ours)	11.18	9.52	48.42	81.03(▲ 3.11)	76.78(▲ 2.94)	90.42(▲ 0.49)	86.12(▲ 0.56)	83.02(▲ 1.67)	78.31(▲ 1.73)
T: RESNET34-UNET [10]	16.30	94.78	26.34	87.09	84.02	91.62	88.83	85.61	82.71
S: ENET [13]	0.35	2.01	57.14	83.40	80.03	90.17	87.73	84.75	81.80
EMKD [4]	0.35	2.01	60.12	84.14(▲ 0.74)	80.92(▲ 0.89)	89.63(▼ 0.54)	86.64(▼ 1.09)	85.80(▲ 1.05)	82.37(▲ 0.57)
CIRKD [5]	0.35	2.01	57.46	84.54(▲ 1.14)	81.06(▲ 1.03)	90.92(▲ 0.75)	87.85(▲ 0.12)	85.28(▲ 0.53)	82.15(▲ 0.35)
CWD [3]	0.35	2.01	53.81	81.62(▼ 1.78)	77.96(▼ 2.07)	89.70(▼ 0.47)	86.19(▼ 1.54)	84.91(▲ 0.16)	81.97(▲ 0.17)
BPKD [6]	0.35	2.01	56.21	82.96(▼ 0.44)	79.33(▼ 0.70)	89.11(▼ 1.06)	86.76(▼ 0.97)	84.97(▲ 0.22)	81.96(▲ 0.16)
CIRGBD(Ours)	0.35	2.01	60.54	85.57(▲ 2.17)	82.23(▲ 2.20)	93.53(▲ 3.36)	90.89(▲ 3.16)	87.00(▲ 2.25)	83.77(▲ 1.97)
T: RESNET34-UNET [10]	16.30	94.78	26.34	87.09	84.02	91.62	88.83	85.61	82.71
S: RESNET18-UNET [10]	10.92	55.31	28.26	85.04	82.10	91.17	88.26	83.27	80.19
EMKD [4]	10.92	55.31	28.52	86.17(▲ 1.13)	83.54(▲ 1.44)	93.56(▲ 2.39)	90.94(▲ 2.68)	85.97(▲ 2.7)	83.13(▲ 2.94)
CIRKD [5]	10.92	55.31	28.62	87.92(▲ 2.88)	85.05(▲ 2.95)	93.76(▲ 2.59)	91.10(▲ 2.84)	86.33(▲ 3.06)	83.38(▲ 3.19)
CWD [3]	10.92	55.31	28.48	86.88(▲ 1.84)	83.79(▲ 1.69)	91.38(▲ 0.21)	88.51(▲ 0.25)	84.01(▲ 0.74)	80.68(▲ 0.49)
BPKD [6]	10.92	55.31	28.57	85.82(▲ 0.64)	82.79(▲ 0.69)	91.81(▲ 0.64)	88.99(▲ 0.73)	84.62(▲ 1.35)	81.90(▲ 1.71)
CIRGBD(Ours)	10.92	55.31	28.73	88.60(▲ 3.56)	85.68(▲ 3.58)	94.05(▲ 2.88)	91.41(▲ 3.15)	87.44(▲ 4.17)	84.62(▲ 4.43)
T: RESNET34-UNET [10]	16.30	94.78	26.34	87.09	84.02	91.62	88.83	85.61	82.71
S: RESNET18 [22]	11.18	9.52	47.62	77.92	73.84	89.93	85.56	81.35	76.58
EMKD [4]	11.18	9.52	46.23	80.14(▲ 2.22)	76.02(▲ 2.18)	89.25(▼ 0.68)	84.76(▼ 0.8)	82.68(▲ 1.33)	78.02(▲ 1.44)
CIRKD [5]	11.18	9.52	46.57	78.77(▲ 0.85)	74.75(▲ 0.91)	89.81(▼ 0.12)	85.50(▼ 0.06)	81.80(▲ 0.45)	77.18(▲ 0.60)
CWD [3]	11.18	9.52	47.12	78.85(▲ 0.93)	74.87(▲ 1.03)	86.39(▼ 3.54)	81.49(▼ 4.07)	81.78(▲ 0.43)	76.76(▲ 0.18)
BPKD [6]	11.18	9.52	45.17	78.45(▲ 0.53)	74.20(▲ 0.36)	89.77(▼ 0.16)	85.09(▼ 0.47)	82.00(▲ 0.65)	77.28(▲ 0.70)
CIRGBD(Ours)	11.18	9.52	47.31	80.16(▲ 2.24)	76.06(▲ 2.22)	90.27(▲ 0.34)	85.90(▲ 0.34)	82.77(▲ 1.42)	77.95(▲ 1.37)

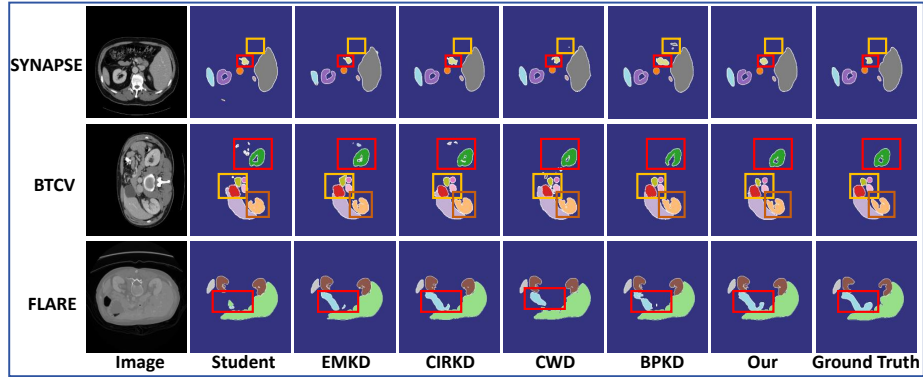


Fig. 4. Segmentation results on three medical datasets are illustrated, where the corresponding visualized student models for Synapse [20], BTCV [8] and FLARE [21] are RESNET18-UNET [10], RESNET18 [22] and ENET [13], respectively. Boxes with the same color indicate comparisons of the same region. It is evident that our segmentation results are significantly better.

4.3 Comparative Experiments

To ensure the fairness and rigor of the comparison, we reimplemented the established knowledge distillation baselines (EMKD [4], CIRKD [5], CWD [3], BPKD [6]), and verified the performance of the proposed CIRGBD method on the BTCV[8], Synapse[20] and FLARE[21] datasets with different lightweight student models (ENET[13], RESNET18[22], RESNET18-UNET[10]) and high-performance teacher models (RA-UNET[12], RESNET34-UNET[10]). As shown in Table 1, CIRGBD achieves the state-of-the-art performance under all experimental configurations, with the improvement of Dice coefficient and mean Intersection over Union (mIoU) no less than 0.34%. The peak improvements reach 4.17% (Dice coefficient) and 4.43% (mIoU), respectively, without performance degradation in any experimental scenario. This result fully validates the effectiveness of the collaborative design of Cross-Image Region Contrastive Distillation (CIRD) and Global Boundary Distillation (GBD) in addressing the core pain points of knowledge distillation for medical image segmentation.

Performance Improvement Analysis. The consistent performance improvement of CIRGBD stems from its targeted optimization for two key challenges in medical imaging. For the high intra-class shape variability and low inter-class contrast of anatomical structures, the CIRD module extracts cross-image category region features in a batch and performs INFOCE contrastive learning, which enhances the intra-class feature similarity and inter-class feature discriminability, enabling lightweight models to accurately identify morphologically variable anatomical structures. This is the core reason why CIRGBD outperforms EMKD (only single-image contrastive learning) as well as CWD (only spatial/channel feature alignment). For example, with ENET as the student model and RA-UNET as the teacher model on the BTCV dataset, CIRGBD boosts the Dice coefficient from 83.40% to 86.53% with an improvement of 3.13%, which is much higher than 1.34% of EMKD and 0.27% of CWD.

For the severe class imbalance caused by the drastic size differences between organs, the GBD module extracts global boundary masks from the teacher model and assigns uniform importance to all organ boundaries, breaking the dominance of large organs in the boundary distillation process and effectively improving the boundary segmentation accuracy of small organs. This solves the inherent defect of BPKD (category-specific boundary alignment, prone to size bias). On the FLARE dataset with extreme organ size differences, taking RESNET18-UNET as the student model and RESNET34-UNET as the teacher model, CIRGBD achieves the highest improvement of 4.17% in Dice coefficient and 4.43% in mIoU, while the Dice coefficient improvement of BPKD is only 1.35%.

Notably, the existing baseline methods often suffer from performance degradation in medical image segmentation scenarios (e.g., CWD and BPKD lead to the decrease of Dice coefficient on the BTCV dataset with RESNET34-UNET as the teacher model and ENET as the student model; EMKD and CIRKD show performance degradation on the Synapse dataset with RA-UNET as the teacher model and RESNET18 as the student model). This is because these methods are designed for natural image segmentation with stable intra-class shape and small

size differences between targets, and lack adaptive optimization for the characteristics of medical images. In contrast, the cross-image region contrast and global balanced boundary distillation of CIRGBD realize the medical-specific knowledge distillation optimization, thus effectively avoiding the above performance instability.

Computational Efficiency. CIRGBD retains the lightweight characteristics of the student model while improving the segmentation accuracy: its parameter count (Param(M)) and floating-point operations (FLOPS(G)) are completely consistent with the baseline methods and the original student models without any additional computational overhead. Moreover, it achieves a faster inference speed (FPS) in most experimental settings (e.g., the FPS of ENET as the student model on the BTCV dataset reaches 57.46, which is higher than 57.12 of EMKD; the FPS of RESNET18 as the student model on the FLARE dataset reaches 48.42, which is higher than 47.86 of BPKD). This advantage comes from the lightweight design of CIRD and GBD modules: both modules only perform simple operations such as mask weighting, average pooling and Sobel edge extraction on the existing feature maps and prediction results of the model, without introducing additional convolution layers or network branches, which is highly in line with the clinical deployment requirements of medical image segmentation and achieves the dual optimization of accuracy and efficiency. This efficient design with no additional overhead fully demonstrates the effectiveness and feasibility of deploying CIRGBD in real-world clinical scenarios.

Qualitative Visualization. As shown in Figure 4, compared with the baseline methods, CIRGBD generates more complete organ segmentation results and smoother boundary delineation. For small anatomical structures such as gallbladder and pancreas in the BTCV dataset, CIRGBD can accurately capture their fine features, which is difficult for the baseline methods to achieve. This visualization result is the direct embodiment of the enhanced category discrimination ability of the CIRD module and the improved boundary segmentation accuracy of the GBD module, and also proves that CIRGBD not only improves the quantitative performance of segmentation metrics, but also generates segmentation results with practical clinical value.

4.4 Ablation Study

Table 2. Ablation experiments on three datasets.

Method	BTCV [8]		Synapse [20]		Flare [21]	
	Dice(%)	mIoU(%)	Dice(%)	mIoU(%)	Dice(%)	mIoU(%)
Teacher: RA-UNET [9,12]	86.17	83.11	92.76	90.03	87.00	84.00
Student: ENET [13]	83.40	80.03	90.17	87.73	84.75	81.80
+ CIRD	84.95(▲ 1.55)	81.65(▲ 1.62)	91.83(▲ 1.66)	88.81(▲ 1.08)	86.18(▲ 1.43)	82.94(▲ 1.14)
+ GBD	85.51(▲ 2.11)	82.15(▲ 2.12)	91.95(▲ 1.78)	88.97(▲ 1.24)	85.46(▲ 0.71)	82.32(▲ 0.52)
+ CIRD + GBD	86.53(▲ 3.13)	83.42(▲ 3.39)	92.27(▲ 2.10)	89.39(▲ 1.66)	86.98(▲ 2.23)	83.97(▲ 2.17)

To validate the effectiveness of the proposed CIRD and GBD modules, and their synergistic effect, we conduct comprehensive ablation experiments on the

BTCV[8], Synapse[20] and FLARE[21] datasets with the teacher model RA-UNET[12] and lightweight student model ENET[13]. All experiments follow the same setup as Section 4.1 to eliminate hyperparameter interference, with the original ENET as the baseline. The quantitative results are presented in Table 2.

Table 2 shows that both CIRD and GBD modules bring consistent performance improvements on all datasets, and their fusion achieves the best performance, which verifies the rationality of the module design and their complementary effect. Specifically, the CIRD module achieves a maximum Dice gain of 1.66% on the Synapse dataset, which verifies the essential role of cross-image region contrastive learning in medical image segmentation. By enhancing intra-class similarity and inter-class discriminability at the feature level, CIRD effectively addresses the problems of high intra-class shape variability and low inter-class contrast of anatomical structures in medical images, making the lightweight model better at identifying morphologically variable organ regions. The GBD module achieves a maximum mIoU gain of 2.12% on the BTCV dataset, which proves the validity of global boundary distillation: by assigning uniform importance to all organ boundaries, GBD mitigates the dominance of large organs in boundary distillation, and significantly improves the boundary segmentation accuracy of small organs, thus optimizing the pixel-level segmentation performance of the model.

Notably, the fusion of CIRD and GBD achieves a peak Dice gain of 3.13% and mIoU gain of 3.39% on the BTCV dataset, with the gain significantly higher than the superposition of single modules, reflecting a strong synergistic effect between the two modules. CIRD improves the discriminability of category semantic features at the intermediate layer, providing an accurate semantic foundation for boundary segmentation; GBD optimizes the global boundary localization accuracy at the prediction layer, making the semantic feature discrimination land in precise pixel-level segmentation. The two modules solve the core pain points of medical image segmentation from the perspectives of *semantic feature representation* and *boundary localization*, respectively, thus realizing the 1+1>2 performance improvement. Moreover, the fused model even surpasses the teacher model RA-UNET on the BTCV dataset (86.53% vs. 86.17% in Dice), which demonstrates that the proposed distillation framework can not only transfer the knowledge of the teacher model to the lightweight student model, but also achieve performance breakthrough through dual optimization of feature and prediction layers.

4.5 Hyperparameter Sensitivity Analysis

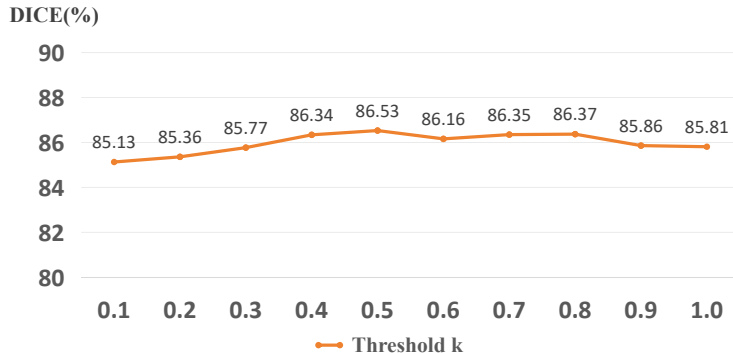
We conduct sensitivity experiments on the hyperparameters α_1 , α_2 , and β in the total loss function to validate the rationality of the selected configuration. Taking the setting where the teacher model is RA-UNET and the student model is ENET on the BTCV dataset as an example, we test the performance under different combinations of these weights, as shown in Table 3.

Table 3. Hyperparameter sensitivity analysis on BTCV (RA-UNET $\rightarrow$ ENET).

α_1	α_2	β	Dice(%)	mIoU(%)
0.1	0.1	0.1	85.12	81.80
0.1	0.5	0.1	86.53	83.42
0.1	1.0	0.1	86.02	82.95
0.5	0.1	0.1	85.41	82.05
0.1	0.5	0.5	86.11	83.01

The results indicate that the optimal configuration $\alpha_1 = 0.1$, $\alpha_2 = 0.5$, and $\beta = 0.1$ achieves the highest Dice and mIoU scores. A larger α_2 consistently outperforms α_1 , confirming that high-level cross-image contrastive learning contributes more to category discrimination. Excessively large weights for auxiliary losses lead to performance degradation, as they may disturb the primary segmentation objective. Therefore, the selected configuration balances the contributions of all components and ensures stable training.

In our experiments, we further tune all hyperparameters to analyze their effects on performance. We observe that the loss weights α_1 , α_2 , β , and the temperature parameter τ have relatively limited influence within a reasonable range, indicating that our method is robust to these parameters. In contrast, the boundary threshold k shows a more significant impact, as shown in Figure 5.

**Fig. 5.** Performance under different boundary threshold k on BTCV. A clear peak is achieved at $k = 0.5$.

Specifically, a larger k only retains pixels with high gradients as boundaries, which produces clean results but misses weak boundaries. A smaller k treats most low-gradient pixels as boundaries, introducing excessive noise. We observe obvious performance drops when k is too small ($k \in \{0.1, 0.2\}$) or too large ($k \in \{0.8, 0.9, 1.0\}$). Empirically, $k = 0.5$ achieves the best trade-off between performance and robustness.

We further visualize the boundary extraction results under $k = 0.3, 0.5, 0.7$ in Figure 6. When k is too small, boundaries are blurred and incomplete. When k is too large, adjacent organs easily fuse. Only $k = 0.5$ maintains clear, complete, and highly discriminable boundaries, so it is chosen as the optimal threshold.

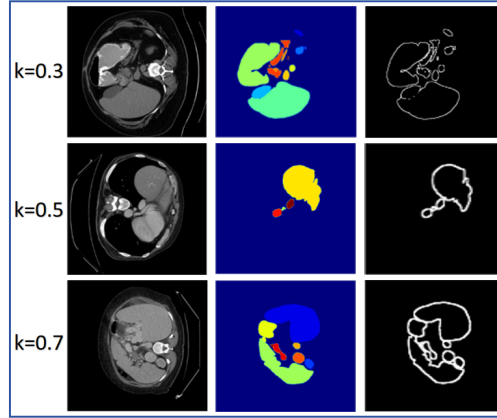


Fig. 6. Qualitative comparison of global boundary extraction with different k . Smaller k leads to blurry boundaries, while larger k causes fusion. The setting $k = 0.5$ yields the best boundary quality.

5 CONCLUSION

This paper proposes two distillation methods for medical image segmentation to address poor category distinguishability and blurry boundaries caused by organ shape diversity and size variation. Specifically, CIRD captures semantic information from intermediate layers to enhance category separability, while GBD extracts boundary information from the output layer to improve edge localization. Extensive experiments on three medical datasets demonstrate the effectiveness of the proposed CIRGBD method.

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